

Chapter 1

Further trigonometry

- Contents:**
- A** Reciprocal trigonometric functions
 - B** Inverse trigonometric functions
 - C** Algebra with trigonometric functions
 - D** Double angle identities
 - E** Compound angle identities

A

RECIPROCAL TRIGONOMETRIC FUNCTIONS

We define the functions $\operatorname{cosec} x$, $\sec x$, and $\cot x$ as the **reciprocal** functions of $\sin x$, $\cos x$, and $\tan x$ respectively:

$$\operatorname{cosec} x = \frac{1}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \text{and} \quad \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$$

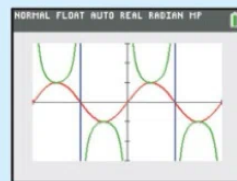
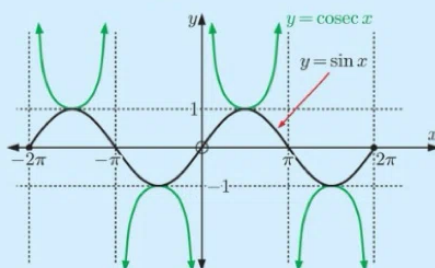
For example, $\operatorname{cosec} \frac{\pi}{6} = \frac{1}{\sin \frac{\pi}{6}} = \frac{1}{\frac{1}{2}} = 2$.

Example 1

Self Tutor

Using the graph of $y = \sin x$, sketch the graph of $y = \frac{1}{\sin x} = \operatorname{cosec} x$ for $-2\pi \leq x \leq 2\pi$. Check your answer using technology.

- The zeros of $y = \sin x$ become vertical asymptotes of $y = \frac{1}{\sin x}$.
- The local maxima of $y = \sin x$ become local minima of $y = \frac{1}{\sin x}$. (1)
- The local minima of $y = \sin x$ become local maxima of $y = \frac{1}{\sin x}$. (-1)
- When $\sin x = 1$, $\frac{1}{\sin x} = 1$ and when $\sin x = -1$, $\frac{1}{\sin x} = -1$.



EXERCISE 1A



1 Without using a calculator, find:

a $\operatorname{cosec} \frac{\pi}{3}$

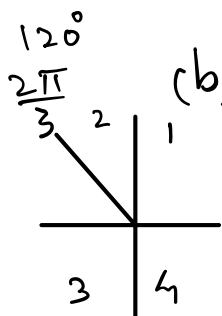
b $\cot \frac{2\pi}{3}$

c $\sec \frac{5\pi}{6} = \frac{1}{\cos(\frac{5\pi}{6})} = \frac{1}{-\frac{\sqrt{3}}{2}} = -\frac{2}{\sqrt{3}}$

d $\cot \pi = \frac{\cos \pi}{\sin \pi} = \frac{-1}{0}$ undefined

(a) $\operatorname{cosec} \frac{\pi}{3} = \frac{1}{\sin \frac{\pi}{3}} = \frac{1}{(\frac{\sqrt{3}}{2})} = \frac{2}{\sqrt{3}}$

(b) $\cot(\frac{2\pi}{3}) = \frac{\cos(\frac{2\pi}{3})}{\sin(\frac{2\pi}{3})} = \frac{(-\frac{1}{2})}{(\frac{\sqrt{3}}{2})} = -\frac{1}{\sqrt{3}}$



Or $\cot(\frac{2\pi}{3}) = \frac{1}{\tan(\frac{2\pi}{3})} = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}$

2 Without using a calculator, find $\operatorname{cosec} x$, $\sec x$, and $\cot x$ given:

1A a $\sin x = \frac{3}{5}$, $0 \leq x \leq \frac{\pi}{2}$

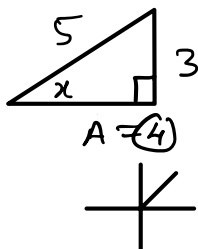
$A = \sqrt{5^2 - 3^2}$
 $= \sqrt{25 - 9}$

$A = \sqrt{16} = 4$

$\therefore \operatorname{cosec} x = \frac{5}{3}$

$\sec x = \frac{H}{A} = \frac{5}{4}$

$\cot x = \frac{A}{O} = \frac{4}{3}$



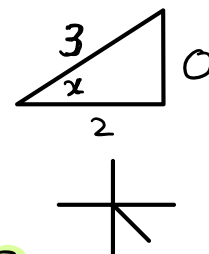
b $\cos x = \frac{2}{3}$, $\frac{3\pi}{2} < x < 2\pi$

$O = \sqrt{3^2 - 2^2}$
 $= \sqrt{9 - 4}$
 $O = \sqrt{5}$

$\therefore \operatorname{cosec} x = \frac{H}{O} = \frac{3}{\sqrt{5}}$

$\sec x = \frac{H}{A} = \frac{3}{2}$

$\cot x = \frac{A}{O} = \frac{2}{\sqrt{5}}$



3 Find the other five trigonometric ratios if:

A a $\cos x = \frac{3}{4}$ and $\frac{3\pi}{2} < x < 2\pi$

c $\sec x = 2\frac{1}{2}$ and $0 < x < \frac{\pi}{2}$

e $\tan \beta = \frac{1}{2}$ and $\pi < \beta < \frac{3\pi}{2}$

b $\sin x = -\frac{2}{3}$ and $\pi < x < \frac{3\pi}{2}$

d $\operatorname{cosec} x = 2$ and $\frac{\pi}{2} < x < \pi$

f $\cot \theta = \frac{4}{3}$ and $\pi < \theta < \frac{3\pi}{2}$.

(A) $\frac{3\pi}{2} < x < 2\pi$

$\cos x = \frac{3}{4}$

$O = \sqrt{4^2 - 3^2}$

$= \sqrt{16 - 9}$

$O = \sqrt{7} \checkmark$

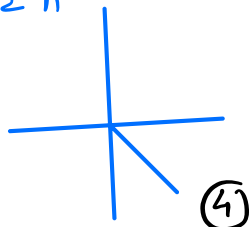
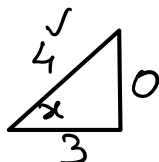
$\therefore \sin x = \frac{O}{H} = -\frac{\sqrt{7}}{4}$

$\tan x = \frac{O}{A} = -\frac{\sqrt{7}}{3}$

$\sec x = \frac{4}{3}$

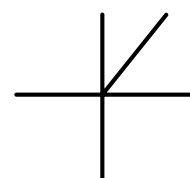
$\operatorname{cosec} x = -\frac{4}{\sqrt{7}}$

$\cot x = -\frac{3}{\sqrt{7}}$



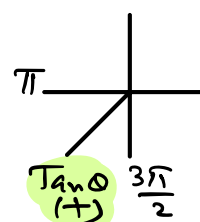
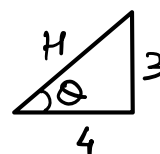
c $\sec x = 2\frac{1}{2}$ and $0 < x < \frac{\pi}{2}$

$\sec x = \frac{5}{2}$



f $\cot \theta = \frac{4}{3}$ and $\pi < \theta < \frac{3\pi}{2}$.

$\tan \theta = \frac{3}{4}$



$H = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = 5$

$\sin \theta = -\frac{3}{5}$

$\operatorname{cosec} \theta = -\frac{5}{3}$

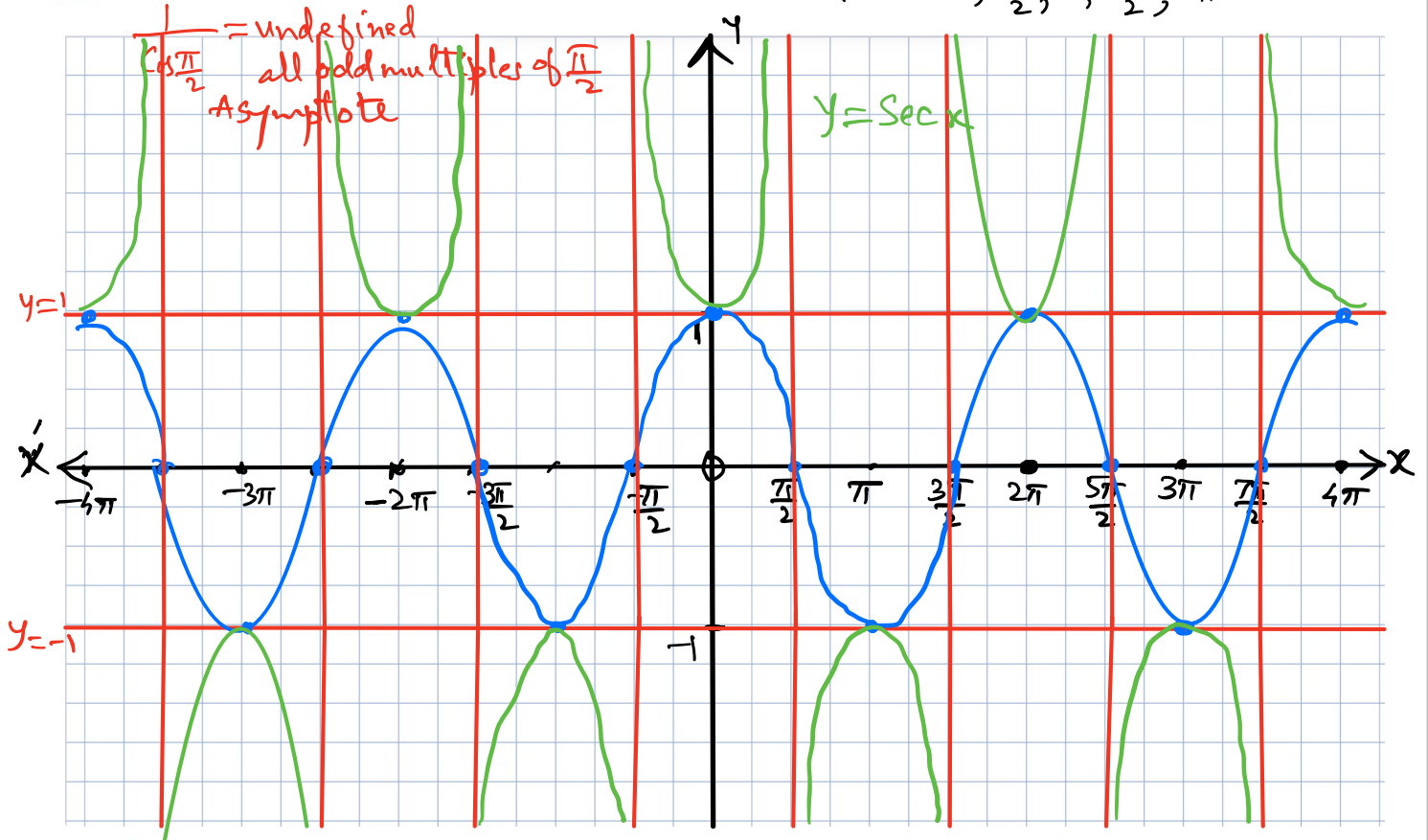
$\cos \theta = -\frac{4}{5}$

$\sec \theta = -\frac{5}{4}$

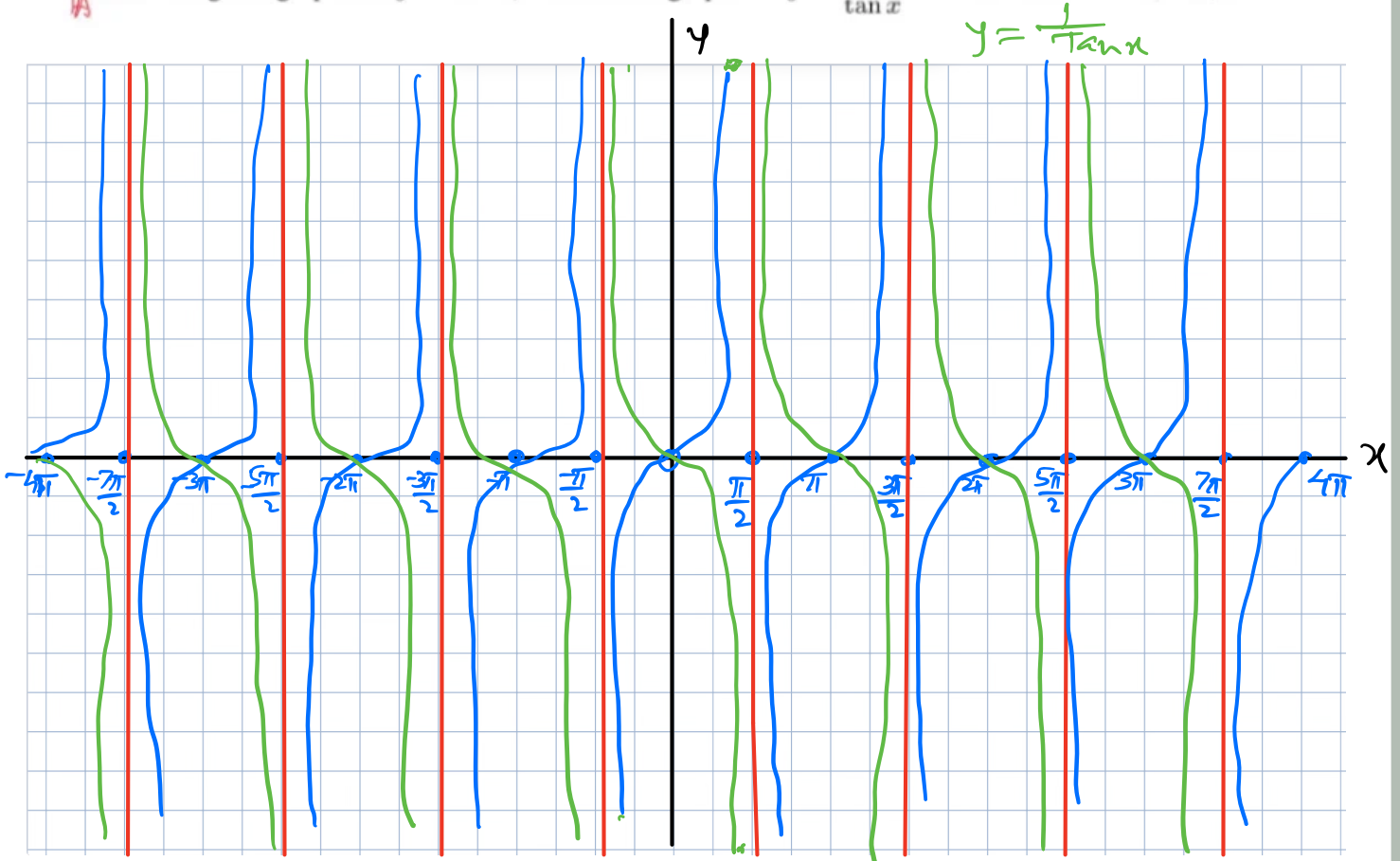
4 Using the graph of $y = \cos x$, sketch the graph of $y = \frac{1}{\cos x} = \sec x$ for $-2\pi \leq x \leq 2\pi$.

1A Check your answer using technology.

$\dots, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \dots$



5 Using the graph of $y = \tan x$, sketch the graph of $y = \frac{1}{\tan x} = \cot x$ for $-2\pi \leq x \leq 2\pi$.



$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{0}{0} \quad \text{at } \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

7 Solve exactly for $0 \leq x \leq 2\pi$:

A a $\sec x = 2$

c $\sqrt{3} \sec 2x = -2$

e $\cot x + 1 = 0$

b $\operatorname{cosec} x = -\sqrt{2}$

d $\operatorname{cosec}(x + \frac{\pi}{6}) + \sqrt{2} = 0$

f $\cot(2x - \frac{\pi}{4}) - \sqrt{3} = 0$

(a) $\sec x = 2$

$\therefore \cos x = \frac{1}{2}$

$x = \frac{\pi}{3}, x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

(b) $\sin x = -\frac{1}{\sqrt{2}}$

$x = \sin^{-1}(-\frac{1}{\sqrt{2}}) = -\frac{\pi}{4}$

$x = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$

$x = \pi + \frac{\pi}{4}$

$= \frac{5\pi}{4}$

(c) $\sqrt{3} \sec 2x = -2$

$\sec 2x = -\frac{2}{\sqrt{3}}$

$\cos 2x = -\frac{\sqrt{3}}{2}$

$\frac{\pi}{6}$



$2x = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ ✓ $0 \leq x \leq 2\pi$

$0 \leq 2x \leq 4\pi$

$2x = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$ ✓

$\therefore 2x = \frac{5\pi}{6} + 2\pi = \frac{17\pi}{6}$

$2x = \frac{7\pi}{6} + 2\pi = \frac{19\pi}{6}$

$\therefore x = \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}$

$2\pi + \frac{\pi}{6}$

d $\operatorname{cosec}(x + \frac{\pi}{6}) + \sqrt{2} = 0$

$0 \leq x \leq 2\pi$

Shift left

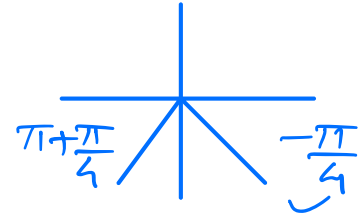
$\operatorname{cosec}(x + \frac{\pi}{6}) = -\sqrt{2}$



$$\sin\left(x + \frac{\pi}{6}\right) = -\frac{1}{\sqrt{2}}$$

$$\frac{\pi}{6} \leq \left(x + \frac{\pi}{6}\right) \leq \frac{13\pi}{6}$$

$$x + \frac{\pi}{6} = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$



$$\& x + \frac{\pi}{6} = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$\therefore x = \frac{7\pi}{4} - \frac{\pi}{6} = \frac{19\pi}{12}$$

$$x = \frac{5\pi}{4} - \frac{\pi}{6} = \frac{13\pi}{12}$$

$$\therefore x = \frac{13\pi}{12}, \frac{19\pi}{12}$$